Scaling symbols and symmetric multilinearisation with the factorial retained

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Status and scope. This is a new proof proposal, not an independent acceptance. It supplies GTZ-LP05 and the scaling assertion E.8(v), including the divisibility used here. Its project input is the exact tautification theorem in [2], still pending independent review. It does not prove the remaining correction units or the inverse theorem. The unpublished erratum motivates the factorial correction but supplies no black-box inference. All source and project pins are in the accompanying contracts. Protocol v1.4's available/needed/connecting-argument distinctions are recorded there.

1 Conventions and preliminary constructions

We use the rational, connected, simply connected nilpotent groups, polynomial maps, finite-dimensional unit-sphere-valued nilcharacters and symbol equivalence of [1, Sections 3–6 and Appendix E]. For a vector-valued function u, $u^{\otimes r}$ means its ordinary tensor power if $r \geq 0$, with $u^{\otimes 0} = 1$, and $\overline{u}^{\otimes(-r)}$ if $r < 0$. Conjugation is always written explicitly. Equivalence at multidegree d means that every component of $u \otimes \overline{v}$ is a nilsequence with support in $\{a : a \leq d,\ a \neq d\}$; all inequalities on multiindices are coordinatewise. In the degree case this is degree at most $d-1$. We use the group law on symbols proved in published E.7 and E.8(i)–(iii). Their proofs use tensor products, conjugation, contraction of a unit vector with its conjugate, and a rational top-group quotient. With tautification, the top group is central, so those proofs apply to the present objects.

Initially all data are ordinary, and all integers d, k, q are fixed standard integers. The last section states exactly how the constructions pass to limits and parameter families. A multidegree nilcharacter on a limit subset comes with a representing global polynomial orbit and test function; we use that representation to extend it, and only then restrict back.

Apply [2, Theorem 1 and Corollary 2] to the given representation. We can work on a taut group and with an exactly equal represented sequence whose polynomial map g satisfies $g(0) = 1$. Positive groups and the vertical character are unchanged. For a degree filtration $G_0 = G_1$; for multidegree G_0 is generated by the G_{e_i}. In particular G_d is central when the support is contained below one positive d. Write $\mathfrak{g}_a = \log G_a$ with its rational structure induced by Γ.

Lemma 1 (Monomial coordinates)**.** *For multidegree support $a \leq d$, the normalised map has an expression*

$$g(n) = \prod_{0<a\leq d} \exp(X_a n^a), \qquad X_a \in \mathfrak{g}_a, \quad n^a = \prod_i n_i^{a_i}. \tag{1}$$

Fix any order refining total weight. For a degree-D map on $\mathbb{Z}^k$, replace $a \leq d$ by $0 < |a| \leq D$ and require $X_a \in \mathfrak{g}_{|a|}$. Every such ordered product is polynomial for the indicated filtration.

Proof. For a degree-D input, use the multidegree target $\mathfrak{g}_a = \mathfrak{g}_{|a|}$: coordinate derivatives are particular degree derivatives, so the input is polynomial for this target. Published Lemma B.9 gives ordered binomial monomials with coefficient at index a in G_a. Expand the finite BCH formula. A product of scalar binomial polynomials with labels $a_1, \dots, a_r$ has monomials n^b only for $b \le a_1 + \cdots + a_r$. Its Lie-bracket coefficient belongs to $\mathfrak{g}_{a_1+\cdots+a_r} \subseteq \mathfrak{g}_b$. Thus $\log g(n) = \sum_{0<b\le d} Y_b n^b$, $Y_b \in \mathfrak{g}_b$; terms outside the support vanish and the constant is zero. To obtain (1), process increasing total monomial weight: choose X_b to match the coefficient at n^b in the BCH logarithm of the partial ordered product. All commutators of previous factors have larger total monomial weight, and their coefficients at b belong to $\mathfrak{g}_b$. Consequently the adjustment is in $\mathfrak{g}_b$ and does not change any previously matched weight. The finite support makes this algorithm terminate. The argument with $\mathfrak{g}_{|b|}$ is identical. For the converse each factor is a polynomial map: in its one-parameter abelian subgroup an iterated difference of n^a vanishes if its coordinate order exceeds a, and otherwise the coefficient stays in $\mathfrak{g}_a \subseteq \mathfrak{g}_b$ at derivative index b. Apply the product theorem [1, Corollary B.4]. This use does not invoke E.8, E.9 or GI. □

We will use rational connected subgroups and rational central quotients. Their generated subgroups and central quotients are the geometric construction proved in [3, Section 2]. We spell out the additional rational-image check used below. For each rational G_a, choose finitely many elements $\gamma_{a,j}$ of $\Gamma \cap G_a$ whose logarithms span $\mathfrak{g}_a$. Such a choice exists: the real span of the logarithms of a nilpotent subgroup is a Lie algebra. Indeed conjugation by integer powers of any of its elements gives a polynomial in that integer, and finite differences extract its iterated brackets; BCH then gives the assertion. Its connected subgroup contains the lattice, so cocompactness forces it to be G_a, by the homogeneous-space argument of [3, Section 2]. The same polynomial/BCH argument gives the rational Lie structure $\operatorname{span}_{\mathbb{Q}} \log \Gamma$; all filtration spaces are rational for this structure.

For the scaling image below, the vectors $(q^{|a|} \log \gamma_{a,j}, \log \gamma_{a,j})$ exponentiate to the actual lattice elements $(\gamma_{a,j}^{q^{|a|}}, \gamma_{a,j})$. For the multilinear image, multiply each $\log \gamma_{a,j}$ by an integer N so that its rational second component in (8) lies in the lattice L constructed next. Its first component then exponentiates to $\gamma_{a,j}^N$. These finitely many lattice elements have logarithms spanning each image Lie algebra $\mathfrak{k}_b$. The finite-generator fact in [1, p. 1243] makes their generated discrete subgroup cocompact in K_b. Thus K_b is rational for the actual product lattice, not merely for an abstract rational form. This also checks $b = 0$.

The following explicit lattice choice records the extra finite-symmetry requirement needed below. In a rational nilpotent Lie algebra with a finite group P of rational automorphisms, choose a full rational lattice L_0 and replace it by $\sum_{\sigma\in P} \sigma L_0$. Clear rational bracket denominators and the finitely many BCH denominators by replacing L_0 by $L = ML_0$. Then $\mathrm{BCH}(L, L) \subseteq L$ and $-L = L$. The group $\exp L$ is discrete. It is cocompact: an adapted rational central-series basis, after integer scaling, lies in L; successive rounding of its coordinates gives a compact set of coset representatives. The same construction makes all specified rational filtration subspaces rational. If a rational linear functional on a specified rational central subspace is prescribed, increase M to make it integral on L intersected with that subspace. This gives a P-invariant lattice, not just a rational structure invariant under P. Each choice uses only fixed geometric data.

2 Scaling and divisibility on the actual orbit

Theorem 2 (Corrected E.8(v))**.** *Let $d \neq 0$, and let $\chi(n) = F(g(n)\Gamma)$ have multidegree at most d. For every standard integer q,*

$$[\chi(qn)] = q^{|d|}[\chi(n)]. \tag{2}$$

For degree $D > 0$ on $\mathbb{Z}^k$, the exponent is D. The equivalence is on the whole domain, hence on every indicated restriction. Its geometric and Lipschitz bounds are independent of the polynomial coefficients.

Proof. Use (1). Form the finite positive Rees Lie algebra

$$\mathcal{R} = \bigoplus_{0<a\le d} \mathfrak{g}_a t^a, \qquad [Xt^a, Yt^b] = [X,Y]t^{a+b},$$

where a component outside the downset is zero. This is a Lie algebra: it is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[t_1, \dots, t_k]$ modulo the ideal spanned by monomials outside the downset. Define the rational Lie map

$$A_q(Xt^a) = (q^{|a|}X, X) \in \mathfrak{g} \oplus \mathfrak{g}.$$

Its bracket identity is exact because $q^{|a|}q^{|b|} = q^{|a+b|}$, including negative and zero q. Let $\mathfrak{k}_b$ be the image of $\bigoplus_{a\ge b,\ a\neq 0} \mathfrak{g}_a t^a$, and put $K_b = \exp \mathfrak{k}_b$. These are rational, descending, and $[\mathfrak{k}_b, \mathfrak{k}_c] \subseteq \mathfrak{k}_{b+c}$. Also $\mathfrak{k}_0 = \sum_i \mathfrak{k}_{e_i}$, so the filtration is taut. The ambient group is K_0, with lattice $K_0 \cap \Gamma^2$. Every factor of

$$(g(qn), g(n)) = \prod_a \exp(n^a A_q(X_a t^a))$$

is polynomial at index a, and hence so is their product. The top group has Lie algebra $\mathfrak{k}_d = \{(q^{|d|}X, X) : X \in \mathfrak{g}_d\}$. For $c = q^{|d|}$, the restricted function

$$(x, y) \longmapsto F(x) \otimes \overline{F(y)^{\otimes c}} \tag{3}$$

is invariant under K_d: the two vertical phases are $c\eta(X)$ and $-c\eta(X)$. This calculation uses $F^{\otimes c}$, with the signed convention above, on the same second coordinate. The rational central quotient K_0/K_d has the quotient filtration with its top group zero. Thus (3) along the displayed orbit is a lower-degree nilsequence, which is precisely (2). Restriction and descent preserve a fixed Lipschitz bound since the geometric maps and compact quotient are fixed. For degree D use $\bigoplus_{0<|a|\le D} \mathfrak{g}_{|a|} t^a$ and the filtration by $|a| \ge j$. The top image is the same graph with exponent D; all other steps agree. At $q = 0$ the first orbit coordinate is 1, and (3) is the constant $F(1)$; no 0^0 or division is used. □

Corollary 3 (Divisibility, with the representative specified)**.** *For positive degree and every standard $Q \ge 1$, there is a nilcharacter ψ on the same domain with $[\chi] = Q[\psi]$.*

Proof. In (1) replace each X_a by $Q^{-|a|}X_a$ and call the resulting map g_Q. It is polynomial on the same group and lattice. With the same F put $\xi(n) = F(g_Q(n)\Gamma)$. Then $g_Q(Qn) = g(n)$ exactly, factor by factor in the same order. Theorem 2 gives $[\chi] = Q^D[\xi]$, where $D = |d|$ or the degree bound, and $\psi = \xi^{\otimes Q^{D-1}}$ gives the asserted equation. The test function still has unit norm; its frequency and finite target dimension change only by the displayed standard tensor power. In particular this is divisibility of the actual symbol, not division of an arbitrarily selected multilinear representative after construction. □

3 Raw multilinearisation and the factorial graph

Let $d = (d_1, \dots, d_k) \neq 0$, $D = |d|$, and divide $\{1, \dots, D\}$ into ordered blocks B_i of size d_i. For a subset J put $w(J) = (|J \cap B_i|)_i$ and $a! = \prod_i a_i!$; set $C = d!$. Construct

$$\widetilde{\mathfrak{g}} = \bigoplus_{\varnothing \neq J \subseteq [D]} \mathfrak{g}_{w(J)} \epsilon_J, \qquad [X\epsilon_J, Y\epsilon_L] = \begin{cases} [X,Y]\epsilon_{J\cup L}, & J \cap L = \varnothing, \\ 0, & J \cap L \neq \varnothing. \end{cases} \tag{4}$$

This is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[\epsilon_1, \dots, \epsilon_D]/(\epsilon_1^2, \dots, \epsilon_D^2)$; Jacobi therefore holds. There is no empty component: normalisation $g(0) = 1$ is what permits its removal. Put

$$\widetilde{\mathfrak{g}}_b = \bigoplus_{1_J \geq b} \mathfrak{g}_{w(J)} \epsilon_J. \tag{5}$$

These are descending rational Lie subalgebras, their bracket has the required index sum, and they vanish when any $b_i > 1$. The zero group is generated by the first-coordinate groups since each nonempty J contains a coordinate. Thus (5) is taut. Its top group is central. The block permutation group $P = \prod_i S_{d_i}$ preserves (4), permutes the groups in (5) by the same permutation of their indices, and acts trivially on the top copy $\mathfrak{g}_d \epsilon_{[D]}$. Choose a P-invariant lattice $\widetilde{\Gamma}$ as in Section 1, making

$$\widetilde{\eta}(\exp(X \epsilon_{[D]})) = \eta(\exp X) \tag{6}$$

integral on its top lattice. This is possible since η is integral on $\Gamma \cap G_d$ and hence rational on the rational top Lie algebra.

Choose a Lipschitz unit-sphere-valued function $\widetilde{F}$ with frequency $\widetilde{\eta}$. Here is the needed existence construction, with no invariance under P assumed. The central torus acts freely on $\widetilde{G}/\widetilde{\Gamma}$: its stabilizer is precisely the top lattice. On finitely many local trivialisations of this torus bundle, take the character (6) times a smooth base bump. Choose the bumps so that their supports cover the compact base and they vanish near the chart boundaries. The vector of these functions has no common zero; divide by its norm. This gives a globally Lipschitz unit vector of the required frequency, with a fixed finite dimension. It is the partition-of-unity construction underlying published (6.3).

For the coefficients of (1), define

$$\widetilde{g}(z) = \prod_{0 < a \leq d} \exp\left(a! \sum_{w(J)=a} z_J X_a \epsilon_J \right), \qquad z_J = \prod_{j \in J} z_j, \qquad \widetilde{\chi}(z) = \widetilde{F}(\widetilde{g}(z)\widetilde{\Gamma}). \tag{7}$$

Each factor is polynomial for (5). To see this without commuting distinct coefficients, fix a; the summands in its exponent commute because their Lie coefficients are the same X_a, so (7)'s factor is a product of the individual monomial factors. Then use product closure. Thus $\widetilde{\chi}$ has multidegree at most $(1, \dots, 1)$. Let Δn repeat n_i throughout B_i.

Lemma 4 (Raw diagonal identity)**.** *The diagonal $\widetilde{\chi}(\Delta n)$ is equivalent at multidegree d to $\chi(n)^{\otimes C}$. In particular for $k = 1$ the multiplier is $d!$.*

Proof. On the positive Rees algebra $\mathcal{R}$ of Section 2, set

$$T(X t^a) = \left(X, a! \sum_{w(J)=a} X \epsilon_J \right) \in \mathfrak{g} \oplus \widetilde{\mathfrak{g}}. \tag{8}$$

This is a rational Lie map. Indeed for each U with $w(U) = a + b$ the number of ordered disjoint decompositions $U = J \sqcup L$ with $w(J) = a, w(L) = b$ is $\prod_i \binom{a_i + b_i}{a_i}$. Multiplying by $a!b!$ gives $(a+b)!$. Thus $[T(Xt^a), T(Yt^b)] = T([X, Y]t^{a+b})$ exactly. If $a + b$ is outside the support, both sides vanish by $[\mathfrak{g}_a, \mathfrak{g}_b] \subseteq \mathfrak{g}_{a+b} = 0$.

Define $\mathfrak{k}_b = T(\bigoplus_{a \geq b, a \neq 0} \mathfrak{g}_a t^a)$, $K_b = \exp \mathfrak{k}_b$. As before these give a descending, rational, taut filtration on $K_0 \subseteq G \times \widetilde{G}$, with lattice $\Lambda = K_0 \cap (\Gamma \times \widetilde{\Gamma})$. They give the actual comparison orbit

$$(g(n), \widetilde{g}(\Delta n)) = \prod_a \exp\left(n^a T(X_a t^a)\right). \tag{9}$$

No asserted equality of top weights is being used to generate a non-descending filtration: every defining sum includes all $a \geq b$. The top is exactly

$$K_d = \{(\exp X, \exp(CX\epsilon_{[D]})) : X \in \mathfrak{g}_d\}. \tag{10}$$

It is rational and central. The function on the comparison nilmanifold is

$$\mathcal{F}(x, y) = F(x)^{\otimes C} \otimes \overline{\widetilde{F}(y)}. \tag{11}$$

Its two vertical phases on (10) are $C\eta(\exp X)$ and $-\widetilde{\eta}(\exp(CX\epsilon_{[D]})) = -C\eta(\exp X)$. Therefore (11) descends to a Lipschitz function F' on K_0/K_d with the projected lattice Λ' and quotient filtration. Writing k'_a for the projected $\exp T(X_a t^a)$ gives the corrected identity

$$\chi(n)^{\otimes C} \otimes \overline{\widetilde{\chi}(\Delta n)} = F'\left(\prod_a (k'_a)^{n^a} \Lambda'\right). \tag{12}$$

The right side has top group zero, so it has support strictly below d. For $k = 1$, (11) contains $F^{\otimes d!}$ and (12) contains $\chi^{\otimes d!}$. Dropping either factor would leave the uncancelled vertical phase $(1 - d!)\eta$ and would not prove the claimed equivalence.

The diagonal itself is a degree-d nilcharacter: coarsen (5) by taking the direct sum of components with $w(J) \geq a$ at index a. This is descending, rational and taut, its top is still the $J = [D]$ component, and (7) on the diagonal is polynomial for it. This verifies the exact symbol domain used in (12), without assuming that arbitrary linear maps preserve arbitrary multidegree filtrations. □

4 Literal symmetry and division before construction

The map $\widetilde{g}$ is equivariant under block permutations: $\widetilde{g}(\sigma z) = \sigma \widetilde{g}(z)$ after fixing the compatible action convention. An arbitrary $\widetilde{F}$ need not be invariant, so equivariance alone does not prove that $\widetilde{\chi}$ is symmetric. We supply the missing transition explicitly.

Let V be the finite-dimensional Hermitian target of $\widetilde{F}$ and $m = |P| = C$. Regard $\operatorname{Sym}^m V$ as the symmetric subspace of $V^{\otimes m}$ with its induced norm, and write $\odot$ for symmetrised tensor product. On the compact nilmanifold put

$$B(x) = \bigodot_{\sigma \in P} \widetilde{F}(\sigma x), \qquad F_{\mathrm{sym}}(x) = B(x)/\|B(x)\|. \tag{13}$$

The product of nonzero vectors in the symmetric algebra is nonzero: after a choice of basis this algebra is a polynomial ring over $\mathbb{C}$, an integral domain. Hence $\|B(x)\| > 0$ everywhere. Continuity and compactness give a positive minimum, so (13) is Lipschitz. Reindexing the finite product proves exact P-invariance. Since P fixes the top group, (13) has vertical frequency $m\widetilde{\eta}$. Consequently

$$\chi_{\mathrm{sym}}(z) = F_{\mathrm{sym}}(\widetilde{g}(z)\widetilde{\Gamma}) \tag{14}$$

is a literally block-symmetric multidegree-$(1, \ldots, 1)$ nilcharacter. On a block diagonal, $\sigma\widetilde{g}(\Delta n) = \widetilde{g}(\Delta n)$. All factors in (13) then equal the same unit vector v, and $v^{\odot m} = v^{\otimes m}$ has norm one. Thus the diagonal of (14) is exactly $\widetilde{\chi}(\Delta n)^{\otimes m}$ in the symmetric subspace, not merely a proportional vector with an uncontrolled scalar. Lemma 4 therefore gives

$$[\chi_{\mathrm{sym}}(\Delta n)] = Cm[\chi(n)]. \tag{15}$$

Theorem 5 (E.9 and E.10 with a symmetric output)**.** *For a multidegree-d nilcharacter χ there is a literally block-symmetric multidegree-$(1,\dots,1)$ nilcharacter $M(\chi)$ whose block diagonal is equivalent to χ. The one-variable case is the asserted symmetric E.9 output. The assertion holds on any given limit subset after restriction.*

Proof. If $d \neq 0$, first use Corollary 3 with $Q = Cm = C^2$ to choose the specified representative ρ with $[\chi] = Cm[\rho]$. Apply (4)–(14) to this ρ, including its actual polynomial coefficients, lattice, function and vertical frequency. Equation (15) now identifies its output diagonal with $[\chi]$. We do not divide the already symmetrised function and assume that its symmetry survives. If $C = 1$, all constructions and division remain valid. Coordinates with $d_i = 0$ have empty blocks; the normalised polynomial is independent of them. If all $d_i = 0$, the original sequence is a constant unit vector and that vector on the zero-variable domain gives the assertion exactly. □

5 The exact input to Section 13

Let $s \geq 2$ be fixed and let $\chi(h, n)$ be the multidegree-$(1, s-1)$ nilcharacter occurring immediately before published (13.4). First choose ρ with $[\chi] = s[\rho]$ by Corollary 3, in that multidegree symbol group. Then set $\widetilde{\chi} = M(\rho)$ from Theorem 5. It has s variables, multidegree $(1,\dots,1)$ and literal symmetry in its last $s-1$ variables. Its actual diagonal gives

$$[\chi(h,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)} = s[\widetilde{\chi}(h,n,\dots,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)}. \tag{16}$$

Here is the comparison of symbol groups in (16). For support $a \leq (1, s-1)$, give the same group the degree filtration generated by the G_a with $|a| \geq j$. Commutators satisfy the index law, rationality follows from the rational generated-subgroup lemma, and the orbit is polynomial by its monomial expansion. At total weight s the only possible index is $(1, s-1)$, so deleting that top index leaves degree at most $s-1$. Thus both multidegree equivalences used above become degree-s equivalences, with the same functions and the same input (h, n). This proves precisely (13.4), including its multiplier s.

The sequence $\Theta(n) = \widetilde{\chi}(n,\dots,n)$ is a degree-at-most-s nilcharacter by the same coarsening with the binary cube. Fixed linear pullbacks $n \mapsto n+h$ preserve that degree filtration and preserve lower-degree equivalence; one may either use the monomial expansion or published Corollary B.5 and E.8(vi) in its degree-only form. No full symmetry in all s variables is asserted here. Section 13's later symmetrisation argument is exactly what must address the first variable. Its uses of multilinearity require the separately proved replacement for Lemma 13.2; that result is not inferred from (16). Likewise the subsequent bias and density arguments remain separate.

6 Uniformity, limit interpretation, and audit boundary

All nilpotent groups, rational subspaces, finite tensor dimensions, integer powers and permutation groups above are fixed by the original geometric data, d, k, and the specified standard divisibility/scaling integers. They do not depend on orbit coefficients or interval length. The bounded-basepoint normalisation supplied by [2] has a uniform Lipschitz constant. Subgroup restrictions, rational quotients and metrics are fixed compact geometric maps. The lattice and the test function in (6)–(7) are selected once from those fixed data. The positive minimum in (13) is for that fixed test function on that fixed compact space, not an infimum over a varying family of independently chosen functions. For a varying original test function with bounded Lipschitz norm, the raw comparison (11) varies, but the auxiliary $\widetilde{F}$ and (13) can still be fixed because their frequency and group are fixed.

Every coefficient change is a finite BCH expression or a rational scalar multiple in a fixed Lie algebra. Apply it coordinatewise to a representing sequence to obtain a limit map. The finite formulas, exact identities, positive fixed lower bound in (13), and uniform Lipschitz estimates pass to the natural ultrapowers used in published Appendix A. Where geometric data range over a standard finite list, perform this construction for each member of that list. No selection depending on an unbounded interval length or uncontrolled pointwise choice is introduced. All vertical identities here have that natural-ultrapower meaning: they hold for every limit point and every limit top-group element, by transfer of the exact ordinary identity. An identity tested only on standard points would not suffice; no such restriction is used here.

The affected-use audit in this unit covers published E.8(v),(vii), E.9, E.10, and (13.4) through the definition of Θ. It also checks the literal symmetry used for the last $s-1$ variables at those sites. It does not assert completeness of all corrections to GTZ. The arbitrary-test-function symmetry transition and the factorial omissions are preserved as findings with proposed repairs awaiting independent mathematical review.

References

[1] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, Annals of Mathematics 176 (2012), 1231–1372. DOI: 10.4007/annals.2012.176.2.11. Authoritative final PDF pinned in contracts.

[2] Project proof, Tautification and the shifted nilcharacter construction, project author theta, tautification-003, Theorem 1 and Corollary 2. Pending independent review.

[3] Project proof, local Lemma 13.2, author theta, lemma13-2-002, Section 2. Pending independent review. Exact version pinned in contracts.